

# Classification\_Assignment 1

## 1.RandomForest:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.95      | 0.91   | 0.93     | 79      |
| 1            | 0.84      | 0.90   | 0.87     | 41      |
| accuracy     |           |        | 0.91     | 120     |
| macro avg    | 0.89      | 0.91   | 0.90     | 120     |
| weighted avg | 0.91      | 0.91   | 0.91     | 120     |

- What is the overall performance of the model in random forest?

Ans: 0.91

- What is the correct classification of not purchased?

Ans: 0.91

- What is the correct classification of purchased?

Ans: 0.90

- Out of all customers predicted as purchased, how many were actually purchased?

Ans: 0.95

- Out of all customers predicted as purchased, how many were actually not purchased?
- Ans: 0.84
- What is the overall balance between precision and recall in predicting purchased?
- Ans: 0.93
- What is the overall balance between precision and recall in predicting purchased?
- Ans: 0.87

## 2. Decision Tree:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.91      | 0.91   | 0.91     | 79      |
| 1            | 0.83      | 0.83   | 0.83     | 41      |
| accuracy     |           |        | 0.88     | 120     |
| macro avg    | 0.87      | 0.87   | 0.87     | 120     |
| weighted avg | 0.88      | 0.88   | 0.88     | 120     |

- What is the overall performance of the model in random forest?

Ans: 0.88

- What is the correct classification of not purchased?

Ans: 0.91

- What is the correct classification of purchased?

Ans: 0.83

- Out of all customers predicted as purchased, how many were actually purchased?

Ans: 0.91

- Out of all customers predicted as purchased, how many were actually not purchased?

Ans: 0.83

- What is the overall balance between precision and recall in predicting purchased?

- Ans: 0.91

- What is the overall balance between precision and recall in predicting purchased?
- Ans: 0.83

### 3.SVM:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.66      | 1.00   | 0.79     | 79      |
| 1            | 0.00      | 0.00   | 0.00     | 41      |
| accuracy     |           |        | 0.66     | 120     |
| macro avg    | 0.33      | 0.50   | 0.40     | 120     |
| weighted avg | 0.43      | 0.66   | 0.52     | 120     |

- What is the overall performance of the model in random forest?
- Ans: 0.66
- What is the correct classification of not purchased?
- Ans: 1.00

- What is the correct classification of purchased?  
Ans: 0.00
- Out of all customers predicted as purchased, how many were actually purchased?  
Ans: 0.66
- Out of all customers predicted as purchased, how many were actually not purchased?  
Ans: 0.00
- What is the overall balance between precision and recall in predicting purchased?
- Ans: 0.79
- What is the overall balance between precision and recall in predicting purchased?
- Ans: 0.00